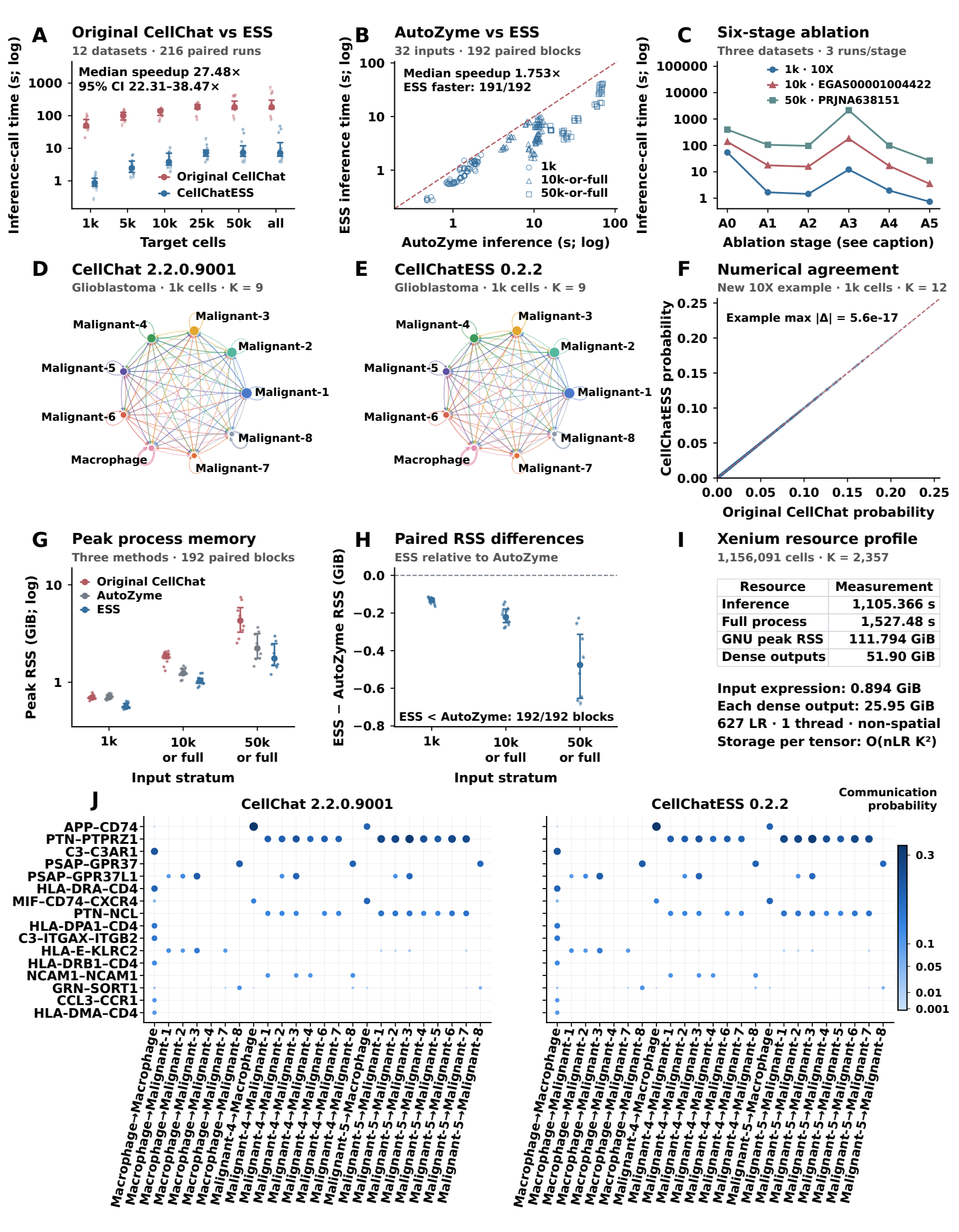

A
Original CellChat vs ESS
12 datasets · 216 paired runs
Median speedup 27.48×
95% CI 22.31–38.47×
Original CellChat
CellChatESS
Inference-call time (s; log)
Target cells
1k
5k
10k
25k
50k
all
B
AutoZyme vs ESS
32 inputs · 192 paired blocks
Median speedup 1.753×
ESS faster: 191/192
1k
10k-or-full
50k-or-full
ESS inference time (s; log)
AutoZyme inference (s; log)
C
Six-stage ablation
Three datasets · 3 runs/stage
1k · 10X
10k · EGAS00001004422
50k · PRJNA638151
Inference-call time (s; log)
Ablation stage (see caption)
A0
A1
A2
A3
A4
A5
D
CellChat 2.2.0.9001
Glioblastoma · 1k cells · K = 9
E
CellChatESS 0.2.2
Glioblastoma · 1k cells · K = 9
Malignant-1
Malignant-2
Malignant-3
Malignant-4
Malignant-5
Malignant-6
Malignant-7
Malignant-8
Macrophage
F
Numerical agreement
New 10X example · 1k cells · K = 12
Example max |Δ| = 5.6e-17
CellChatESS probability
Original CellChat probability
G
Peak process memory
Three methods · 192 paired blocks
Original CellChat
AutoZyme
ESS
Peak RSS (GiB; log)
Input stratum
1k
10k or full
50k or full
H
Paired RSS differences
ESS relative to AutoZyme
ESS − AutoZyme RSS (GiB)
ESS < AutoZyme: 192/192 blocks
Input stratum
I
Xenium resource profile
1,156,091 cells · K = 2,357
Resource | Measurement
Inference | 1,105.366 s
Full process | 1,527.48 s
GNU peak RSS | 111.794 GiB
Dense outputs | 51.90 GiB
Input expression: 0.894 GiB
Each dense output: 25.95 GiB
627 LR · 1 thread · non-spatial
Storage per tensor: O(nLR K²)
J
CellChat 2.2.0.9001
CellChatESS 0.2.2
Communication probability
APP-CD74
PTN-PTPRZ1
C3-C3AR1
PSAP-GPR37
PSAP-GPR37L1
HLA-DRA-CD4
MIF-CD74-CXCR4
PTN-NCL
HLA-DPA1-CD4
C3-ITGAX-ITGB2
HLA-E-KLRC2
HLA-DRB1-CD4
NCAM1-NCAM1
GRN-SORT1
CCL3-CCR1
HLA-DMA-CD4
Macrophage→Macrophage
Macrophage→Malignant-1
Macrophage→Malignant-2
Macrophage→Malignant-3
Macrophage→Malignant-4
Macrophage→Malignant-7
Macrophage→Malignant-8
Malignant-4→Macrophage
Malignant-4→Malignant-1
Malignant-4→Malignant-2
Malignant-4→Malignant-3
Malignant-4→Malignant-4
Malignant-4→Malignant-6
Malignant-4→Malignant-7
Malignant-4→Malignant-8
Malignant-5→Macrophage
Malignant-5→Malignant-1
Malignant-5→Malignant-2
Malignant-5→Malignant-3
Malignant-5→Malignant-4
Malignant-5→Malignant-5
Malignant-5→Malignant-6
Malignant-5→Malignant-7
Malignant-5→Malignant-8